

Advanced RAG with Hard Negative Mining for Multi-Hop Question Answering

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Abstract. Retrieval-Augmented Generation (RAG) systems ground language model outputs in retrieved evidence, reducing hallucination in open-domain question answering. A critical bottleneck is the retriever: dense retrievers require hard negative training examples to distinguish superficially similar but factually incorrect passages. We build a complete RAG pipeline on HotpotQA, training a Dense Passage Retrieval (DPR) model with BM25-mined hard negatives, indexing passages in a FAISS vector database, and integrating Flan-T5-xl as a generative reader. Hard negative mining improves Recall@1 by 13.0 percentage points over vanilla DPR (0.844 vs. 0.714) and by 14.0 points over BM25 (0.704), with consistent gains in MRR@10 (0.895 vs. 0.793) and nDCG@10 (0.811 vs. 0.688 for vanilla DPR). End-to-end evaluation with Flan-T5-xl achieves F1=0.583 and Exact Match=0.466. An ablation study over three retrieval configurations shows that the pretrained (Natural Questions) DPR does not consistently outperform the lexical BM25 baseline — it leads marginally at R@1 but falls behind at R@3, R@5, and R@10 — and that the decisive improvement is driven entirely by hard negative fine-tuning, and is statistically significant ($p < 0.001$, paired bootstrap). A per-category breakdown identifies bridge (multi-hop) questions as the dominant remaining failure mode, while yes/no questions are handled well.

Keywords: Retrieval-Augmented Generation · Dense Passage Retrieval · Hard Negative Mining · Multi-Hop Question Answering · HotpotQA

1 Introduction

Large language models (LLMs) are prone to hallucination when answering factual questions from parametric memory alone [4]. Retrieval-Augmented Generation (RAG) addresses this by conditioning answer generation on passages retrieved from an external corpus, grounding outputs in verifiable evidence. The retrieval component is a critical bottleneck: if the retriever fails to surface the correct passage, no reader can recover the correct response regardless of its quality.

Classical lexical retrievers such as BM25 [7] fail on paraphrased or semantically complex queries. Dense Passage Retrieval [1] overcomes this by learning semantic representations via bi-encoder models, but requires carefully constructed training data. Randomly sampled negatives provide insufficient gradient signal [2]; hard negative mining selects passages that score highly on a retrieval metric yet do not contain the gold answer, forcing the retriever to learn fine-grained semantic distinctions.

Multi-hop question answering tasks such as HotpotQA [5] require combining evidence from two supporting passages, making them especially sensitive to retrieval quality. In this paper we: (1) mine hard negatives using BM25, (2) fine-tune DPR with a contrastive objective, (3) index all passages in a FAISS vector database, and (4) integrate Flan-T5-xl [9] as a generative reader. We conduct a systematic ablation study and provide a qualitative error analysis of failure cases.

2 Related Work

Karpukhin et al. [1] introduced DPR, a bi-encoder model matching questions and passages via dot-product similarity. Trained with in-batch negatives on Natural Questions, DPR substantially outperformed BM25 and established the foundation for dense retrieval. Our work extends this architecture with domain-specific hard negatives on HotpotQA.

Xiong et al. [2] formalised negative quality in contrastive training through ANCE, which refreshes hard negatives from the retriever’s own top results at each checkpoint. Their analysis demonstrated that random negatives produce trivially separable training signal. We adopt their core insight using BM25-retrieved passages as hard negatives — a computationally cheaper alternative that ANCE itself showed to be competitive with retriever-based mining.

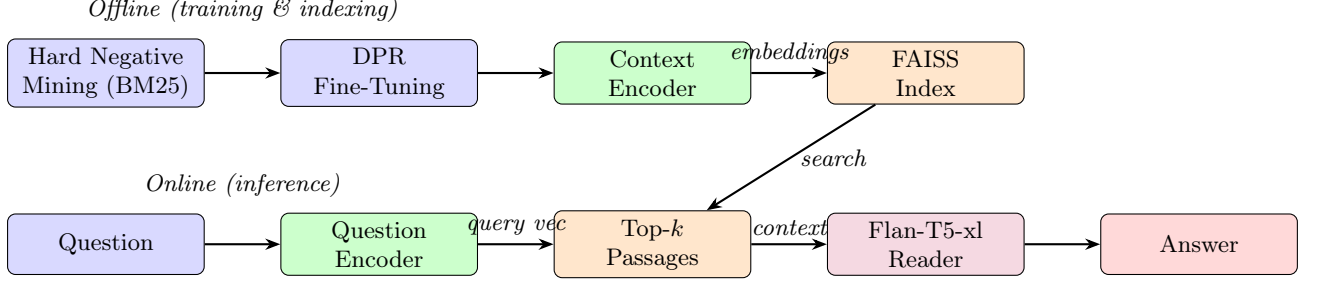


Fig. 1. End-to-end RAG pipeline. The offline stage mines hard negatives with BM25, fine-tunes DPR, and uses the context encoder to build a FAISS index over the corpus. At inference, the question encoder retrieves top- k passages which Flan-T5-xl reads to generate the final answer.

Qu et al. [3] further improved dense retrieval through cross-encoder-filtered hard negatives in RocketQA, achieving state-of-the-art retrieval and providing an upper bound for hard negative training. Lewis et al. [4] proposed the RAG framework motivating our pipeline, showing that grounding generation in retrieved passages substantially reduces hallucination. Yang et al. [5] introduced HotpotQA, which requires aggregating evidence from two supporting passages and has become the standard multi-hop reasoning benchmark. Khattab and Zaharia [6] proposed ColBERT’s late-interaction retrieval, which achieves higher accuracy than standard bi-encoders at the cost of larger index size and serves as a reference for understanding retrieval quality trade-offs. Izacard and Grave [10] introduced Fusion-in-Decoder (FiD), which encodes retrieved passages independently with a T5 encoder and fuses them in the decoder, directly motivating our choice of a generative reader that can synthesise information across multiple passages. Our Flan-T5-xl reader follows the same encode-and-generate paradigm without requiring task-specific reader fine-tuning.

3 System Overview

Figure 1 illustrates the end-to-end pipeline. Hard negative mining and DPR fine-tuning occur offline; the context encoder is also used offline to encode all corpus passages and build the FAISS index. At inference time, only the question encoder, the pre-built FAISS index, and the Flan-T5-xl reader are active.

4 Methodology

4.1 Hard Negative Mining

For each training question q_i with gold passage set $\mathcal{G}_i$, we run BM25 over the 10 candidate passages and select passages that score highly but do not belong to the gold supporting facts: $\mathcal{H}_i = \{p \in \mathcal{P} \setminus \mathcal{G}_i \mid \text{rank}_{\text{BM25}}(p, q_i) \leq K\}$, with $K = 5$. These hard negatives are lexically similar to the question, making them genuinely difficult to distinguish from true supporting passages. In practice, each question yields *at most* 5 hard negatives: examples for which no gold passage can be identified are discarded, and the available distractor pool may be smaller than K for some questions. We mine 30,000 examples from the HotpotQA train split; after discarding unanswerable examples and reserving 20% as a held-out diagnostic set, approximately 24,000 examples remain for training.

4.2 DPR Fine-Tuning

We fine-tune the pretrained DPR question and context encoders (`facebook/dpr-*-single-nq-base`) with contrastive loss. For a mini-batch of size B , each question’s positive passage is placed on the diagonal of the similarity matrix; all remaining in-batch positives *together with all hard negatives* serve as negatives. Concretely, the passage matrix $\mathbf{P}$ has $B + B \cdot n_{\text{HN}}$ rows, where n_{HN} is the number of hard negatives per question in the current epoch:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(f(q_i)^\top f(p_i^+))}{\sum_{j=1}^{B+B \cdot n_{\text{HN}}} \exp(f(q_i)^\top f(p_j))} \quad (1)$$

where $f(\cdot)$ is the encoder’s pooler output (768 dimensions) and the denominator ranges over all B positives plus $B \cdot n_{\text{HN}}$ hard negatives concatenated in the batch. We use AdamW with $lr = 2 \times 10^{-5}$, gradient clipping (`max_norm = 1.0`), and a linear LR scheduler with 10% warmup steps to stabilise contrastive training. We apply a **curriculum strategy**: epoch 1 uses 1 hard negative per question, epoch 2 uses 3, and epoch 3 uses the full 5. This prevents early gradient instability caused by excessively difficult negatives at the start of training. We checkpoint every epoch and select the final model by **global-corpus retrieval** on the ablation set, not by the per-question mini-pool used during training: with only the ~ 7 in-batch candidates per question, the mini-pool metric saturates and correlates poorly with open-corpus retrieval. The final model (*DPR + HN*) is trained on the full $\sim 24,000$ -example training split with batch size 16.

4.3 Training Configuration

Table 1 lists the hyperparameters used for DPR fine-tuning. Training uses in-batch negatives together with curriculum hard negatives, gradient clipping, and a warmup-decay learning-rate schedule throughout.

Table 1. DPR fine-tuning hyperparameters.

Hyperparameter	Value
Base model	facebook/dpr-*-single-nq-base
Optimizer	AdamW
Learning rate	2×10^{-5}
LR scheduler	Linear warmup (10%) + linear decay
Batch size	16
Epochs	3
Max question length	64 tokens
Max passage length	256 tokens
Hard negatives/question	Curriculum: 1 $\rightarrow$ 3 $\rightarrow$ 5 per epoch
Gradient clipping	<code>max_norm = 1.0</code>
Checkpoint selection	Best epoch by global-corpus retrieval
Hardware	NVIDIA A100 (40 GB)

4.4 FAISS Vector Index

All unique passages from the 500 validation examples (approximately 4,900 passages after title-based deduplication) are encoded offline with the context encoder and indexed using FAISS `IndexFlatIP` [8], which performs exact inner-product search. At inference time, the question encoder produces a query vector and FAISS returns the top- k nearest passages by dot-product similarity. This global index simulates a realistic retrieval scenario where the retriever searches a shared corpus rather than the per-question local pool of 10 candidates, making evaluation significantly harder.

4.5 Generative Reader

The top-3 retrieved passages are concatenated (truncated to 2,000 characters) and passed to Flan-T5-xl [9] (3B parameters) with an instruction prompt: “*Answer the question based on the context below. Context: [passages]. Question: [q]. Answer:*” The model generates the answer with greedy decoding (up to 50 tokens) via `model.generate`. We use a generative reader because multi-hop HotpotQA answers are frequently yes/no responses or short inferential spans that benefit from synthesising evidence across passages rather than copying a single contiguous span.

5 Experimental Setup

We evaluate on the first 500 examples of the HotpotQA distractor validation split [5]. Each question has 10 candidate passages (2 gold + 8 distractors). We report **Recall@k** (fraction of questions for which at least one gold passage appears in the top- k retrieved passages, $k \in \{1, 3, 5, 10\}$), **MRR@10** and **nDCG@10** (ranking quality, treating the supporting-fact passages as relevant), **Token F1** (token-level overlap after

Table 2. Retrieval quality on 500 HotpotQA validation examples over the same global FAISS corpus. Recall@k is the fraction of questions with at least one gold passage in the top k ; MRR@10 and nDCG@10 measure ranking quality.

Configuration	R@1	R@3	R@5	R@10	MRR@10	nDCG@10
BM25	0.704	0.862	0.904	0.980	0.793	0.692
DPR (vanilla)	0.714	0.854	0.888	0.932	0.793	0.688
DPR + HN	0.844	0.936	0.962	0.984	0.895	0.811

Table 3. End-to-end QA on 500 HotpotQA validation examples (Flan-T5-xl reader on the top-3 retrieved passages). Better retrieval propagates to higher answer quality.

Configuration	F1	EM
BM25	0.496	0.388
DPR (vanilla)	0.485	0.384
DPR + HN	0.583	0.466

lowercasing and punctuation removal), and **Exact Match (EM)** (normalised string equality). The complete codebase is available at https://github.com/ahmet-yilmaz-cs/CENG467_Term_Project.

We compare three configurations under identical conditions (same 500 validation examples, same global passage corpus of approximately 4,900 passages, same Flan-T5-xl reader). BM25 searches this corpus with `rank_bm25`; both DPR variants build a FAISS index over the same passages: **(1) BM25** — lexical retrieval with `rank_bm25`; **(2) DPR (vanilla)** — pretrained DPR without fine-tuning; **(3) DPR + HN** — DPR fine-tuned on the hard-negative training split.

6 Results

6.1 Ablation Study

Table 2 reports retrieval quality and Table 3 the end-to-end QA metrics across all three configurations. **Effect of hard negative mining.** DPR + HN improves Recall@1 by 13.0 percentage points over vanilla DPR (0.844 vs. 0.714) and by 14.0 points over BM25 (0.704). The gains hold at every cutoff (R@3 +8.2 pp, R@5 +7.4 pp over vanilla) and, crucially, at the top of the ranking: MRR@10 rises from 0.793 to 0.895 and nDCG@10 from 0.688 to 0.811. Because the reader consumes only the top-3 passages, these top-of-ranking gains are exactly what propagate to answer quality. The results confirm that hard negative mining substantially strengthens retrieval by forcing the model to distinguish semantically similar but factually incorrect passages.

BM25 vs. vanilla DPR. The pretrained DPR (Natural Questions) does not consistently outperform the lexical BM25 baseline: it leads marginally at R@1 (0.714 vs. 0.704, a difference of one percentage point), but falls behind at R@3, R@5, and R@10 (e.g. 0.932 vs. 0.980 at R@10) and is essentially tied at MRR@10 and nDCG@10. This pattern is expected — vanilla DPR has not been adapted to HotpotQA’s multi-hop bridging structure — and shows that out-of-the-box dense retrieval does not reliably improve over lexical search. The decisive advantage appears only after hard negative fine-tuning, which surpasses both baselines decisively across all metrics.

6.2 End-to-End QA Performance

The full pipeline (DPR + HN + FAISS + Flan-T5-xl) achieves F1 = 0.583 and EM = 0.466 (Table 3). The same reader on the vanilla DPR and BM25 retrievers reaches only F1 = 0.485 and F1 = 0.496 respectively, so the retrieval gains from hard negative mining translate directly into a ~ 0.10 F1 and 0.08 EM improvement downstream.

Reader input budget. The single largest reader-side lever was the tokeniser input limit. At 512 tokens, the instruction, the 2,000-character context, and the question together exceed the budget and the question is truncated; raising the limit to 1,024 tokens recovers roughly 4 F1 points uniformly across all configurations (DPR + HN: 0.539 $\rightarrow$ 0.583 F1). This was a larger gain than any prompt or model change we tried.

Reader comparison. Table 4 compares three readers on the fixed DPR + HN retriever, each in its best configuration. Flan-T5-xl (3B, encoder-decoder) slightly outperforms the larger Qwen2.5-7B-Instruct [11]

Table 4. Reader comparison on the fixed DPR + HN retriever (top-3 passages, 500 validation questions), each reader in its best configuration. The 7B and 3.8B decoder-only models are 4-bit quantised.

Reader	F1	EM
Flan-T5-xl (3B)	0.583	0.466
Qwen2.5-7B-Instruct	0.574	0.452
Phi-4-mini-Instruct (3.8B)	0.520	0.390

Table 5. DPR + HN end-to-end performance by question category (500 validation questions; the dev split is entirely “hard” level).

Category	n	F1	EM	R@1
Bridge	404	0.535	0.416	0.817
Comparison	96	0.781	0.677	0.958
Yes/no	33	0.909	0.909	0.970
Span	467	0.559	0.435	0.835

and the smaller Phi-4-mini-Instruct [12]. Few-shot prompting with answer post-processing helps the decoder-only instruction model (Qwen: +0.025 F1) but is neutral for the others, and the long multi-part instruction actually *harms* Flan-T5-xl — prompt sensitivity is model-specific. The reader runs zero-shot; its absolute score is further bounded by multi-hop questions that require synthesising both supporting passages, analysed next.

Reader fine-tuning (separate experiment). To isolate the retrieval contribution, the main ablation deliberately uses a fixed, zero-shot reader: it is a neutral instrument, identical across all configurations, and it shows that the retrieval gains transfer to a general-purpose reader without benchmark-specific training. As a separate experiment, we LoRA fine-tune Flan-T5-xl on the HotpotQA *train* split only (8,000 examples, gold-passage context) and evaluate it on the same validation questions with the DPR + HN retriever. Fine-tuning raises end-to-end performance only modestly — F1 0.583 $\rightarrow$ 0.613, EM 0.466 $\rightarrow$ 0.492 — whereas the retrieval improvement contributed roughly +0.10 F1. This confirms that the reader is not the bottleneck: retrieval is. We therefore report reader fine-tuning here rather than folding it into the main ablation.

6.3 Statistical Significance and Breakdown

Significance. On the 500 validation questions, DPR + HN improves F1 by +0.097 (95% bootstrap CI [+0.061, +0.134]) and Recall@1 by +0.130 ([+0.086, +0.172]) over vanilla DPR; both gains are significant at $p < 0.001$ (paired bootstrap, 5,000 resamples). The headline metrics with 95% CIs are DPR + HN R@1 = 0.844 [0.812, 0.876] and F1 = 0.582 [0.542, 0.621], confirming the improvements are not artefacts of the sample size.

Strict (both-passage) recall. HotpotQA answers depend on *two* supporting passages. Under the stricter criterion that *both* gold passages appear in the top 5, recall drops sharply — DPR + HN 0.962 $\rightarrow$ 0.616, vanilla 0.888 $\rightarrow$ 0.462, BM25 0.904 $\rightarrow$ 0.406 — revealing the true difficulty of multi-hop retrieval that the standard “at least one gold” criterion masks. DPR + HN retains the largest margin (+0.15 over vanilla).

Per-category breakdown. Table 5 breaks DPR + HN end-to-end performance down by question type. Comparison questions (F1 0.781) are handled far better than bridge questions (F1 0.535), which require following a bridge entity across two passages and are the dominant failure mode. Yes/no questions, though few (33/500), are answered nearly perfectly (EM 0.909) — contrary to the common assumption that generative readers mishandle them.

7 Error Analysis

We inspected failure cases from the DPR + HN pipeline to identify recurring error patterns, categorising them into retrieval failures (gold passage not in top-3) and reader failures (gold passage retrieved but answer wrong).

Retrieval failures — entity disambiguation. The retriever fails on questions that require precise disambiguation of similarly-named entities. For example:

Q: Who was known by his stage name Aladin and helped organisations improve their performance as a consultant? Gold answer: Eenasul Fateh

BM25 and vanilla DPR both retrieve passages about the fictional character Aladdin (high lexical overlap with “Aladin”), missing the passage about Eenasul Fateh, a Bangladeshi-British artist. DPR + HN correctly surfaces the gold passage by leveraging semantic context around the consultant relation, demonstrating the value of hard negative training.

Yes/no questions are handled well. Contrary to the common assumption that generative readers mishandle binary questions, the breakdown (Table 5) shows yes/no is in fact the strongest category (EM 0.909). Occasional formatting slips still occur:

Q: Are the Laleli Mosque and Esmâ Sultan Mansion in the same neighbourhood? Gold: no Predicted: They are in different neighbourhoods of Istanbul

Here the prediction is semantically correct but fails exact match; such cases are rare, and a concise-answer instruction largely resolves them.

Multi-hop reasoning failures. Some questions require chaining two facts where the connection is implicit:

Q: The director of the romantic comedy “Big Stone Gap” is based in what New York neighbourhood? Gold: Greenwich Village, New York City Predicted: New York City

The reader identifies the correct city but omits the specific neighbourhood, yielding partial F1 and EM = 0. Quantitatively, bridge questions — which require this chaining — score far below comparison questions (F1 0.535 vs. 0.781; Table 5), making them the dominant failure mode.

Context truncation failures. The reader input is truncated to 2,000 characters to fit within the model’s effective context window. For questions whose gold evidence appears late in the retrieved passages, the truncation discards the relevant sentences entirely:

Q: Which film has the director born first, “All Roads Lead to Rome” or “Amreeka”? Gold: Amreeka Predicted: All Roads Lead to Rome

The passage containing the director’s birth year for *Amreeka* falls beyond the 2,000-character cutoff, causing the reader to generate an answer based on incomplete evidence. Increasing the context budget (e.g., using a longer-context reader such as Flan-T5-xl with extended positional encoding) would directly address this failure mode.

8 Discussion

Hard negative mining is the decisive factor. The jump from vanilla DPR (R@1 0.714, roughly on par with BM25 at 0.704) to DPR + HN (R@1 0.844) validates our core hypothesis: the gain comes from domain-specific hard negatives, not from dense retrieval per se. Training on the full ~24,000-example split with gradient clipping, a warmup-decay schedule, and curriculum hard negatives is stable and improves monotonically across epochs, with no sign of degradation at scale.

Checkpoint selection must match the evaluation setting. During training we score checkpoints on the per-question mini-pool (~7 candidates: the positives plus their hard negatives), but this metric saturates and does not track open-corpus quality. Selecting the final epoch on the global corpus instead is what reliably identifies the best model; relying on the mini-pool metric can select a checkpoint that looks strong locally yet retrieves poorly at corpus scale. This also explains the gap between a near-saturated training-time Recall@5 and the global-corpus Recall@5 = 0.962 reported here — the two measure fundamentally different retrieval difficulties.

Reader limitations. The Flan-T5-xl reader runs zero-shot. The dominant remaining failure mode is bridge (multi-hop) questions (F1 0.535 vs. 0.781 for comparison), followed by context truncation; yes/no and comparison questions are already handled well. A stronger or fine-tuned reader and a larger context budget would partially address the remaining cases.

9 Ethical Considerations

Hallucination risk. Flan-T5-xl is a generative model and may produce answers not grounded in retrieved passages. We mitigate this by constraining input to retrieved evidence and using greedy decoding; however, the model can still hallucinate when retrieved passages are insufficient or misleading.

Coverage bias. HotpotQA is derived from Wikipedia, which overrepresents English-language and Western-centric content. Retrieval and reading performance is expected to be lower on questions about underrepresented topics or named entities with limited Wikipedia coverage.

Evaluation limitations. Token F1 and Exact Match against a single reference answer penalise semantically correct paraphrases. For yes/no questions, a model generating a correct explanatory sentence receives partial F1 credit despite being factually correct, distorting aggregate evaluation.

10 Conclusion

We presented a complete RAG pipeline for multi-hop question answering combining BM25-based hard negative mining, DPR fine-tuning, FAISS vector indexing, and Flan-T5-xl generative reading. Hard negative mining improves Recall@1 by 13.0 points over vanilla DPR and 14.0 points over BM25, with matching gains in MRR@10 and nDCG@10, and these retrieval gains carry through to the end-to-end QA metrics (F1 = 0.583, EM = 0.466). Our ablation shows that out-of-the-box dense retrieval does not consistently outperform BM25 on this multi-hop benchmark — leading only marginally at R@1 while falling behind at higher cutoffs — and that the improvement is attributable entirely to hard negative fine-tuning. Selecting checkpoints on the global corpus rather than a small per-question pool is essential for a faithful comparison. Future work should scale hard negative mining to the full 90k training split, add a cross-encoder reranking stage to lift both-passages recall, and evaluate over a full open-domain Wikipedia corpus for an even more realistic retrieval scenario.

References

1. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.: Dense passage retrieval for open-domain question answering. In: Proceedings of EMNLP, pp. 6769–6781 (2020)
2. Xiong, L., Xiong, C., Li, Y., Tang, K.F., Liu, J., Bennett, P., Ahmed, J., Overwijk, A.: Approximate nearest neighbor negative contrastive learning for dense text retrieval. In: Proceedings of ICLR (2021)
3. Qu, Y., et al.: RocketQA: An optimized training approach to dense passage retrieval for open-domain question answering. In: Proceedings of NAACL, pp. 5835–5847 (2021)
4. Lewis, P., et al.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Proceedings of NeurIPS, pp. 9459–9474 (2020)
5. Yang, Z., et al.: HotpotQA: A dataset for diverse, explainable multi-hop question answering. In: Proceedings of EMNLP, pp. 2369–2380 (2018)
6. Khattab, O., Zaharia, M.: ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In: Proceedings of SIGIR, pp. 39–48 (2020)
7. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval* 3(4), 333–389 (2009)
8. Johnson, J., Douze, M., Jégou, H.: Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data* 7(3), 535–547 (2021)
9. Chung, H.W., et al.: Scaling instruction-finetuned language models. arXiv:2210.11416 (2022)
10. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open domain question answering. In: Proceedings of EACL, pp. 874–880 (2021)
11. Qwen Team: Qwen2.5 Technical Report. arXiv preprint arXiv:2412.13663 (2024)
12. Microsoft: Phi-4 Technical Report. arXiv preprint arXiv:2411.15124 (2024)